INCH-POUND

MIL-G-50680C (AR)
30 APRIL 90
SUPERSEDING
MIL-G-50680B (AR)
26 **November 1984**

MILITARY SPECIFICATION

GATE CIRCUIT, MULTIPLE BOARD ASSEMBLY - UPPER: 9342753

This specification is approved for use within the U.S. Army Armament, Munitions and chemical command, and is available for use by all Departments and Agencies of the Department of Defense.

1. **SCOPE**

1.1 Scope. **This specification covers the Board** Assembly - Upper (9342753), hereafter referred to as upper board assembly, which is used in Multiple Gate Circuit (9342754), a subassembly of Radar Receiver R-1475/VPS-2, part. of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for

MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

MIL-S-45743 Soldering, Manual Type, High Reliability Electrical and Electronic Equipment
MIL-S-46844 Solder Bath Soldering of Printed Wiring Assemblies
MIL-G-50678 Multiple Gate, Circuits 9342754
MIL-A-55339/17 - Adapter, Connector

STANDARDS

MILITARY

MIL-STD-109 **Quality** Assurance Terms and Definitions
MIL-STD-810 Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, **PA 19111-5094**

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be these in effect on the date of the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9342753 - Board Assembly - Upper
9342754 - Multiple Gate, Circuits
9342756 - Schematic, Multiple Gate Circuit
10548179 - Divider, 4 Way
10548183 - Hybrid Junction

PACKAGING DATA SHEET

PS9342753 - packaging of Board Assembly - Upper

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein (except **for associated-detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.**

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9342753, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the upper board assembly **shall meet the requirements of MIL-F-13926 as follows:**

a. **Dimensions and tolerances**
b. **Inorganic protective surface finishings**
c. **Part identification and marking**
d. **Workmanship**

3.3 **Environmental service conditions.**

.3.3.1 Vibration. The upper board assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually **perpendicular axes, as follows:**

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall **be 120 minutes per axis.**
The sweep rate **shall be 3.75 minutes ± l0% per decade.**
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 **Shock. The upper board assembly shall not suffer damage and shall meet requirements of 3.4 and 3.5 after exposure to a total of 18** impulse shocks consisting of three shock impulses of **15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.**

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:33Z
Check the source to verify that this is the current version before use.

3.3.3 Altitude.

3.3.3.1 Operating. The upper board **assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).**

3.3.3.2 Non-operating. The upper board **assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).**

3.3.4 Temperature.

303.4.1 Operating. The upper board assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 deg.F and +155 deg. F.

3.3.4.2 Non-operating (storage). The upper board assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60 deg.. **F** for a period of 24 hours and +160 deg. F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The upper board assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. The circuitry shall be in accordance with electrical schematic Drawing 9342756. The symbols referenced in subsequent paragraphs pertain to this schematic and are used herein.

3.5 Performance. Prior to installing the Government Furnished Equipment **(GFE)** Multiple Gate 9342754 in the test setup, replace the Upper Board Assembly 9342753 of the GFE unit with the upper board assembly selected for test.

3.5.1 Inputs. With the selected upper board assembly installed in the GFE unit, construct the test setup per Figure 1 with the following applied inputs:

+15.0 ± O.lVdc to J1l-1, Return to J1l-3.
-15.0 ± O.lVdc to J1l-5, Return to J1l-3.
20KHZ ± 1% pulse repetition frequency, 2.5 ± O.1Vdc amplitude, 2.00 ± 0.01 microseconds (us) pulse width at J1.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:33Z
Check the source to verify that this is the current version before use.

3.5.2 Outputs.

3.5.2.1 Gate offsets. With range gate input (JI) on unit under test open circuited.

3.5.2.1.1 Sum gate. Adjust R39 on the unit under test to produce a voltage at terminal 4 of T2 that is identical (± lOmV) to the voltage at terminal 5 of T2 but opposite in polarity.

3.5.2.1.2 Late gate. Adjust R57 on the unit under test to produce a voltage at Terminal 4 of T3 that is identical (± l0mV) to the voltage at terminal 5 of T3 but opposite in polarity.

3,5.2.1.3 Early--gate. Adjust R70 on the unit under test to produce a voltage--at terminal 4 of T4 that is identical (± lOnl"Vj to . the voltage at terminal 5 of T4 but opposite in polarity.

3.5.2.2 **Early/late gate tuning. With the multiple gate connected with the test equipment as shown in Fire 2, a 60 ± 1% MHZ signal and a monitored 2.8V p-p signal per gate applied to J5, J7, and J10 of the unit under test, the total burst width shall be 750 ± 50 nanoseconds with the notch at the 90% point not** more than 80 nanoseconds. There shall be no dead time between gates at room ambient. The base line between the early/late gate shall not exceed 0.02 microseconds over temperature range without readjust-**ment from ambient setting.**

3.5.2.3 Gate burst.

3.5.2.3.1 Sum gate. With the multiple gate connected to the test equipment as shown in Figure 1, a 60 ± 1% MHZ signal and **a** monitored 2.8V p-p signal per gate applied to J5, J7, and J10 of the unit under test, the gated burst at J4, SUM GATE, shall be:

Peak-to-peak amplitude	2.2V minimum
Width	75 ± ●05us
Rise time	Less than .05 us
Fall time	Less than .05 us
Tilt	Less than or equal to O.lV
Isolation	E on / E off greater than 79.4 ratio

3.5.2.3.2 Late gate. The gated burst output at J6, LATE GATE, shall be per 3.5.2.3.1 except the pulse width shall be 370 ± 25 nanoseconds and the tilt not more than 0.05v.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:33Z
Check the source to verify that this is the current version before use.

3.5.2.3.3 Early gate. The gated burst output at JO, EARLY GATE, shall be per 3.5.2.3.2

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Workmanship. The workmanship and soldering requirements of MIL-F-13926, MIL-S-45743 and MIL-S-46844 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become **a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance** comply **with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective** material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection requirements specified herein are classified **as follows:**

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	**Temperature**	**+50 deg to +90 deg F.**
b.	**Relative humidity**	**90% RH Max.**
c.	**pressure**	**28 to 32 inches of Hg.**

4*3 **First article inspection.**

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three upper board assemblies for inspection in **accordance with** 4.3.2.

4.3.2 Inspections to be performed. First **article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (Functional Test) and Table** II (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. **If any assembly, component or test specimen** fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with **any of the stated requirements.**

4.4 **Quality conformance inspection.**

4.4.1 Inspection lot formation. The term "inspection lot" is defined as **a** homogeneous collection **of** units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply **with the homogeneity criteria specified herein,** regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. **All components,** assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 **Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated** classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first **article approval and prior to Government acceptance or rejection by item or lot.**

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE 1. Functional tests (100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	**N/A**
Major:			
101.	**Electrical**	**3.4**	4.7.2
102.	**Performance**	**3.5**	4.7.3
Minor:		**None**	N/A

4.5 Special sampling. One upper board assembly out of the **first 50 produced and one out of each** 100 produced thereafter shall **be selected at random. The sample shall have met the requirements and tests of Table 1 and shall then meet the tests in Table II. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

TABLE II. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
103.	**Vibration**	3.3.1	4.7.1*1
104.	**shock**	3.3.2	4.7.1.2
105.	**Altitude**	3.3.3	4.7.1.3
106.	**Temperature**	3.3.4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5
Minor:		None	N/A

4.5.1 Failure of sample. should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in **accordance with** the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides **for** Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 **Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the 'Methods **of** Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test **equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.**

4.6.2.3 Test equipment. **In** conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III shall be used in the performance of the applicable test methods and procedures as **specified in 4.7.**

TABLE III. Test equipment list

ITEM NUMBER		DESCRIPTION
1.	**Phase** shifter	**Model** PS-3-60, Merrimac or equal.
2.	Signal generator	60 MHz with variable output from 100 mV to 1.5 V. (HP 606 or equal) .
3.	Pulse generator	PRF: 20 KHz. Amlitude: 2.5 **Pulse width: 2.0 us. Synchronizing output capable of being advanced from O to 2.0 us ahead of output pulse. (Data Pulse 101 or equal).**
4.	Oscilloscope	**Bandwidth: O to 60 MHz.Sensitivity: 5** rev/cm to **0.5 V/cm.** Time base: **0.1** us to **1.0** us/cm with external synchronization. (Tektronix 454 or equal) .

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

TABLE III. Test equipment list (continued)

ITEM NUMBER	DESCRIPTION
5. **power supply**	**+15.0** Vdc and **-15.0** Vdc power source 200 mA. Line/load regulation less than 0.1%. Ripple: less than 1 mv. (Trygon DL 40-1, dual or equal).
6. **Test harness**	**See Figures 1 and 2. With exception of wires connecting the power supplies and unit under test, all cables shall be Radio Frequency type RG-58/U with connectors compatible with mating connectors on test equipment and** unit under test (MIL-A-55339/17).
7. **Hybrid junction**	**P/N 10548183 (1).**
8. **Switch**	**DPST, Cutler hammer P/N 8370R8 or equal.**
9. Divider, 4 way	10548179
10. Digital voltmeter	Fluke 8600 or 8100A or equal.
11. Connector (Pi)	10549074-2
12. Connector (TPS-"T")	Automatic Metals G4768-000-000 or equal.
13. Connector (BNC-"T")	M55239\17-00174(2) .
140 Termination, 50-Ohm	Texscan TF 50 or equal(4).
15. Adapter	BNC female to TPS male(2).

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-81O as **modified by 3.3.1 herein; In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The upper board assembly shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the upper board assembly shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.**

4.7.1.2 Shock. This test **shall be** performed as specified in Procedure I, Section II of Method **516.3 of MIL-STD-81O except that:**

a. The post test functional check (Step 5 of Procedure 1) need *not* be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the upper board assembly shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating) This test shall be Performed as specified in Procedures II and III, Method **500.2** of **MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.**

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be **performed as specified in Procedures I and II, Method 501.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.**

4.7.1.4.2 Low **(operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

4.7.1.5 Humidity (non-operating) . This test shall be performed in accordance with Procedure III, **Method 507.2 of** MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Conditions of operation. The GFE multiple gate shall be used as the test bed. Prior to substituting the upper board assembly of the GFE unit with the upper board assembly selected for testing, the GFE multiple gate shall have met the requirements of MIL-G-50678 (AR).

The test equipment shall be setup as per Figure 1. with switch S1, **(Table III, item 8) off, adjust the +15v power supply (Table III, item 5) for +15.0 ± 0.1 Vdc and the -15V power supply (Table III, item 5) for -15.0 ± 0.1 Vdc using the digital voltmeter (Table III, item 10). Remove the rf generator cable.**

Turn S1 of the test setup to on and recheck the ± 15V settings. Connect a cable from the TPS "T" on the input to J1 of the unit **under test to the oscilloscope (Table III, item 4) input. Disconnect the 50-ohm termination (Table III, item 14)** from the oscilloscope, **and adjust the pulse generator (Table III, item 3), for the following: prf 20 ± 1% KHZ, +2.5 ± O.lV amplitude and a width of 2.0 ± 0.01 microseconds.**

Disconnect the oscilloscope lead from the TPS "T" connector (Table 111, item 12) at J1.

4.7,3.2 Gate offset. Disconnect lead from pulse generator **(Table III, item 3) to J1 of the unit under test.**

4.7.3.2.1 **Sum gate. Using the digital** voltmeter (Table III, item 10) measure the voltage present on terminal 5 of T2 of the unit under test. Adjust R39 to produce an opposite polarity identical (± l0mV) voltage on terminal 4 of T2.

4.7. 3.2.2 Late gate. using the digital voltmeter (Table III, item 10) measure the voltage present on" terminal 5 of T3 in unit under test. Adjust R57 to produce an opposite polarity identical **(± lOmV)** voltage-on **terminal 4 of T3.**

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:33Z
Check the source to verify that this is the current version before use.

4.7.3.2.3 Early gate. Using the digital voltmeter present on terminal 5 of T4 in unit under test, adjust R70 to produce an opposite polarity identical (± l0mV) **voltage on terminal 4 of T4**.

4.7.3.3 Early/late **gate adjustment.**

4.7.3.3.1 Late gate. Connect the unit under test to the test setup per Figure 2, except terminate J3 and J4 of the 4 way divider (Table 111, item 9) with 50-ohm terminations (Table III, item 14) and connect J5 of the 4 way divider to the BNC "T" connector (Table III, item 13) on channel 1 of the oscilloscope. Adjust rf generator to 60 ± 1% MHZ, and for 2.8V p-p on channels 1 and 2 of the oscilloscope.

4.7.3.3.2 Early gate. Connect **the unit under test to the test setup per Figure 2. Turn switch S1 to on. Adjust the phase shifter (Table III, item 1) for minimum disturbance thru gate time and adjust rf generator to maintain 2.8V p-p. The total burst as seen on channel 1 of the oscilloscope (Table 111, item 4) shall be as specified in 3.5.2.2. See Figure 3. If required, adjust us on unit under test to meet specifications. Turn switch S1 to off.**

4.7.3.4 Gate burst. Disconnect lead from pulse generator (Table III, item 3) to J1 of the unit under test.

4.7.3.4.1 Sum gate. Connect the unit under test to the test setup **per Figure 1. Recheck 4 way divider (Table III, item 9) J2 setting. Turn switch S1 to ON and recheck the ± 15V setting. Connect the oscilloscope lead to J4 (sum gate output). The 60 MHZ gated burst shall be per 3.5.2.3.1. See Figure 4.**

4.7.3.4.2 Late gate. Connect the oscilloscope lead to J6 (late gate output) on the unit under test. Connect the 50-ohm termination to J4. The 60 MHZ gated burst shall be per 3.5.2.3.2. See Figure 4.

4.7.3.4.3 **Early gate. Connect the oscilloscope lead to J8 (early gate output) on the unit under test. Connect the 5O-ohm termination to J6. The 60 MHZ gated burst shall be per 3.5.2.3.3. See Figure 4. Turn switch S1 to OFF.**

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:33Z
Check the source to verify that this is the current version before use.

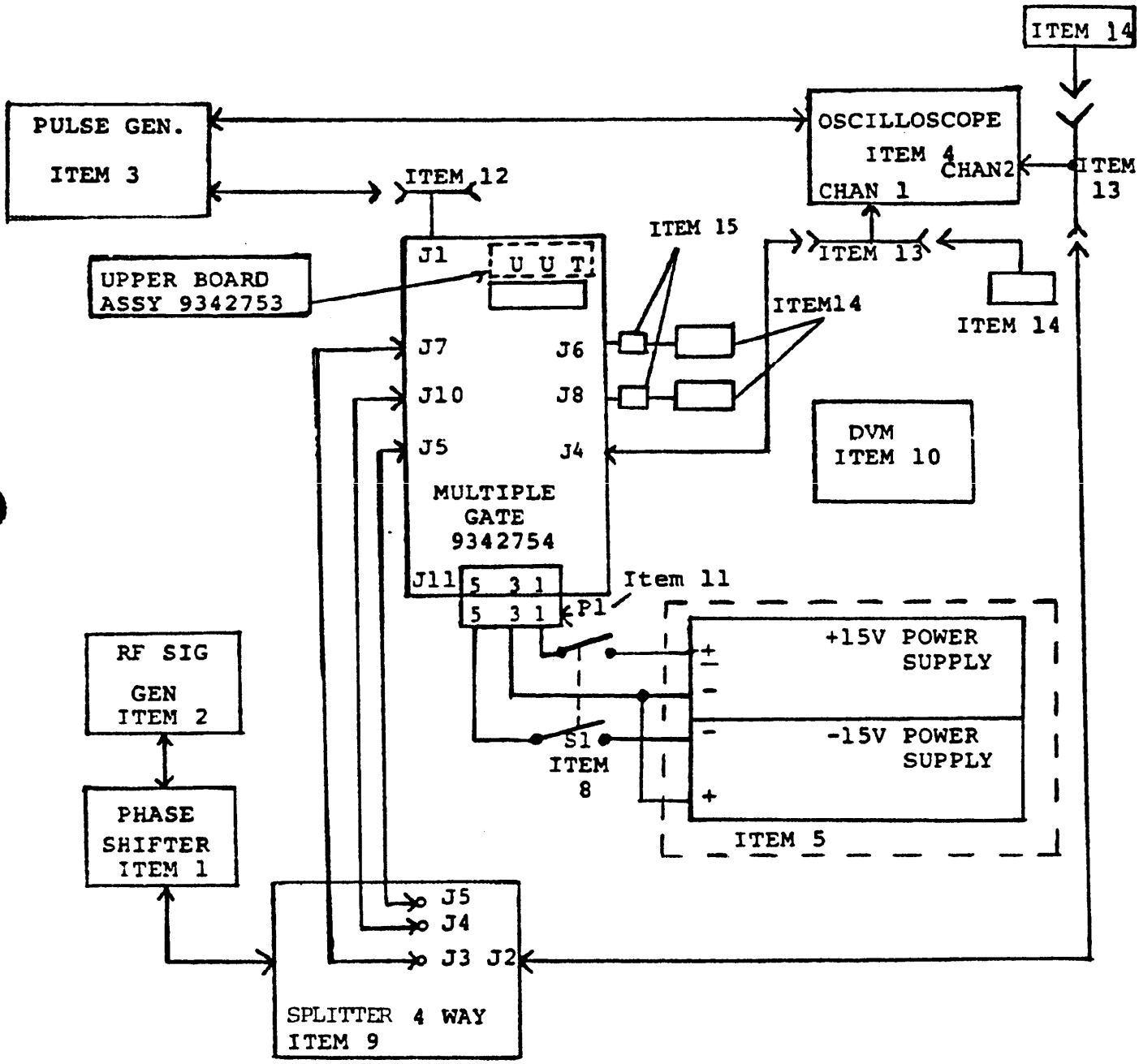


FIGURE 1. Multiple gate circuit test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

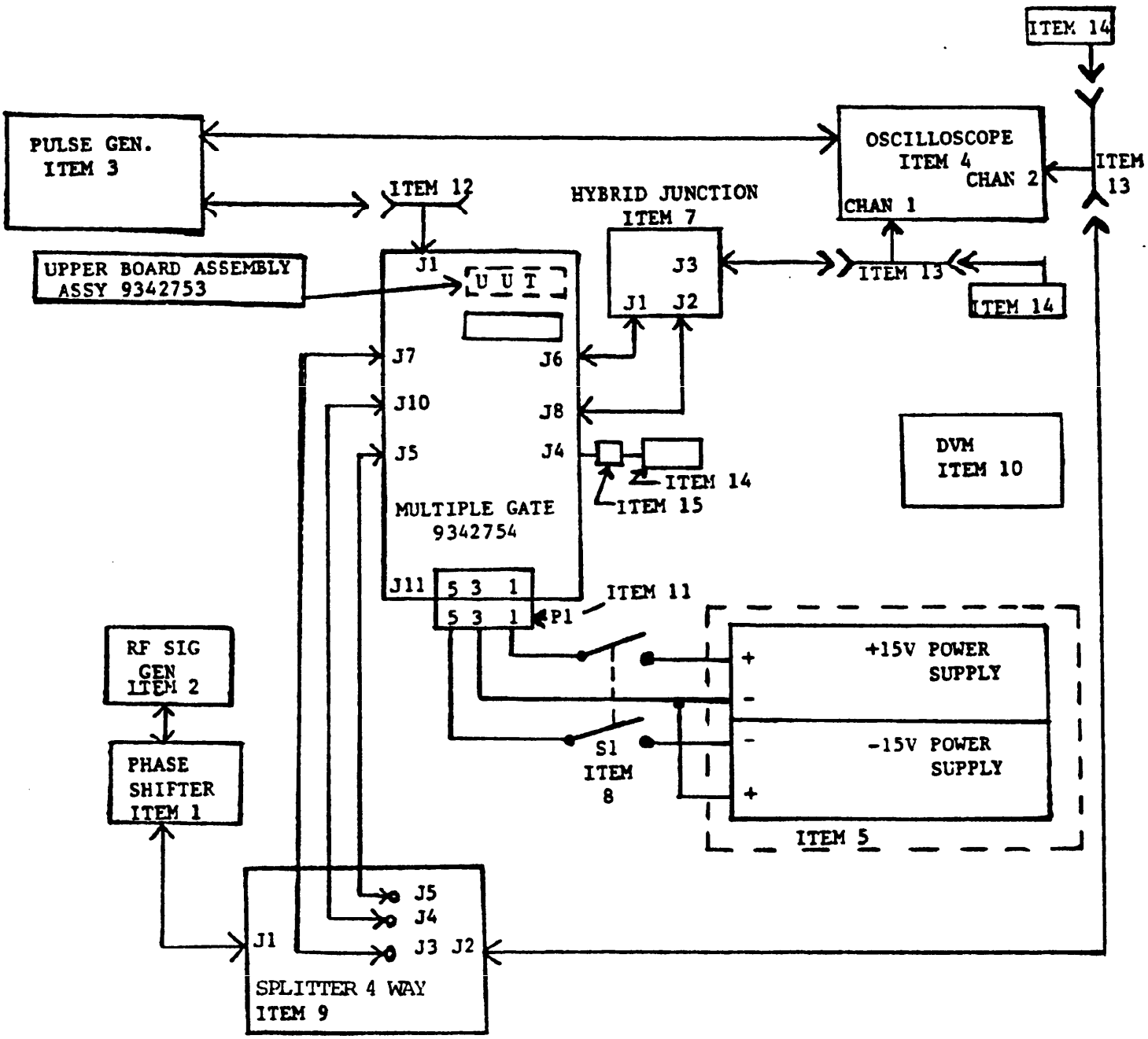


FIGURE 2. Multiple gate circuit test setup.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

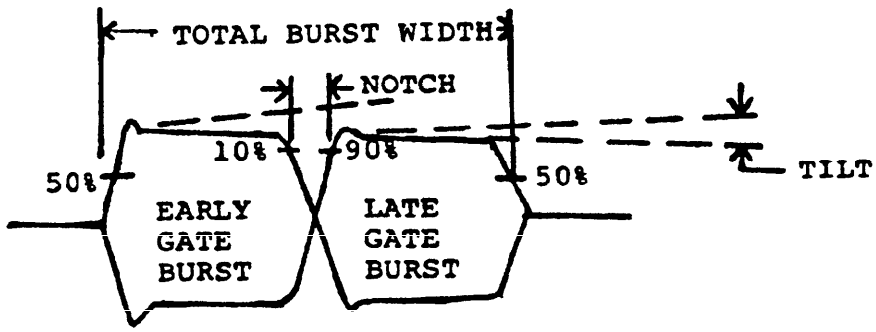


FIGURE 3. Gate outputs

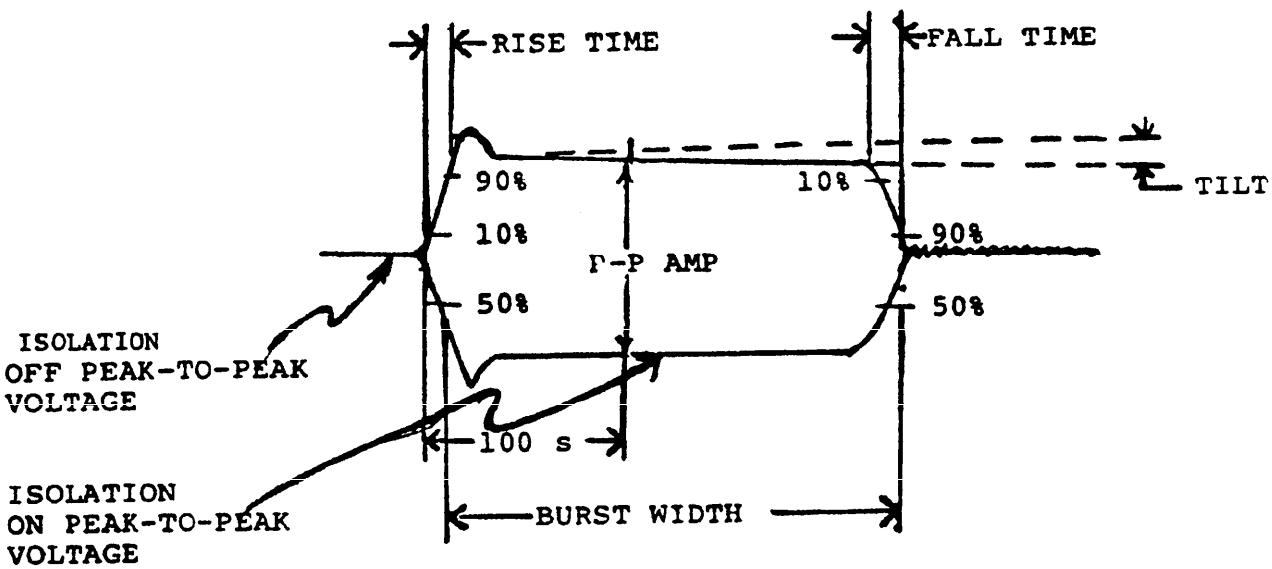


FIGURE 4. Gated burst

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance packaging Data Sheet 9342753.

6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Board **Assembly - Upper is used in the multiple gate circuits assembly, 9342754, a subassembly of Radar Receiver R-1475/VPS-2, part of Radar Set AN/VPS-2. The multiple gate circuits are used to generate three 60 MHZ gated outputs, the** sum, early, and late gates.

6.2 Acquistion requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

Title, **number and date of this specification.**
b. **Applicable stock number.**
c. **First Article Sample requirements.**
d. **packaging requirements, if other than specified in Section 5.**
e. **Serialization requirements, if applicable**

6.3 Submission of contractor inspection equipment designs **for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN : AMSMC-QAF-I , Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.** Unless otherwise specified, **data item DI-R-1714 will apply.**

6.4 Drawings. Drawings listed in Section 2 **of this** specification under the heading U.S. **Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.**

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

6.5 Subject **term** [Key word) listing.

Upper board assembly
Multiple gate circuits
Radar Receiver R-1475/VPS-2
Radar Set AN/VPS-2
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A715)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-50680C (AR)	30 APRIL 1990

3. DOCUMENT TITLE
GATE CIRCUIT, MULTIPLE BOARD ASSEMBLY - UPPER: 9342753

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US ARMY ARDEC STANDARDIZATION & SPECIFICATION OFFICE	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-EAC-S/BLDG 6 PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2020-10-22T20:33Z
Check the source to verify that this is the current version before use.